

REGRESSION ANALYSIS REPORT

Prepared By:

Jeffrey Flores

Prepared For:

Professor Wu,

STA 4155 CMWA.

TABLE OF CONTENTS

I. Introduction.....	1
-----------------------------	----------

II.	Study 1.....	1
	A. Model Summary & Interpretation.....	1
	B. Regression Equation	2
	C. Correlation Test & Visualization	3
	D. Model Assumptions	4
III.	Study 2	5
	A. Model Summary & Interpretation.....	5
	B. Regression Equation.....	8
	C. Model Selection Techniques.....	9
	D. Multicollinearity.....	11
	E. Model Assumptions.....	13
IV.	Conclusion.....	15

I. Introduction

The purpose of this project is to analyze two distinct datasets using regression techniques. In Study 1, the goal is to examine the relationship between the amount of fire damage (measured in thousands of dollars) and the distance (measured in miles) from the fire to the nearest fire station using a simple linear regression model. In Study 2, I focused on the Salaries dataset, where the objective is to construct the most optimal multi-linear regression model. The model aims to explore the association between the response variable, Salary, and ten potential predictor variables, which include both quantitative and qualitative factors.

II. Study 1

A. Model Summary & Interpretation

```
Call:
lm(formula = Damage ~ Distance)

Residuals:
    Min       1Q   Median       3Q      Max
-3.4682 -1.4705 -0.1311  1.7915  3.3915

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  10.2779     1.4203   7.237 6.59e-06 ***
Distance      4.9193     0.3927  12.525 1.25e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.316 on 13 degrees of freedom
Multiple R-squared:  0.9235,    Adjusted R-squared:  0.9176
F-statistic: 156.9 on 1 and 13 DF,  p-value: 1.248e-08
```

Our primary variables of focus for this study are fire damage (measured in thousands of dollars) and distance (measured in miles). Therefore, to examine the relationship between the two, a simple linear regression was conducted using Distance as the predictor variable and Damage as the response variable. The regression model summary showed an R-squared of 92.35%. In the context of the data, this means that 92.35% of the variability in damage can be

explained by distance. This suggests a very strong model fit. When looking at the p-value, you can see that it is extremely small, a number close to 0. A low p-value signifies that the variable is statistically significant, meaning a relationship does exist.

The residual standard error (RSE) was 2.316. To interpret this number, we need to understand that the response variable, Damage, is in thousands of dollars, so 2.316 can be interpreted as \$2,316. Looking at this number at face value does not signify anything, which is why you also need to first know the median amount of damage in the dataset, was 26.10 or \$26,100. Given that the mean damage in the dataset is \$26,100, the model's prediction error is about 8.9%, which indicates a fairly accurate model for a simple regression.

B. Regression Equation

Once I saw all the important indicators of a very strong model were met, I then turned to forming the equation, which came out to be **Damage = 10.2279 + 4.9193 × Distance**. The slope (4.9193) signifies that for every additional mile between the fire and the nearest fire station, the expected fire damage increases by approximately \$4,919. This suggests that greater distance leads to significantly more damage, which is understandable as it would take longer for help to arrive, therefore leading to a longer duration for the house to be on fire. Its standard error was 0.3927, which means that we can expect that the estimate for every additional mile will vary by \$393 in repeated samples. Since the SE is relatively small compared to the coefficient, this indicates a reliable estimate. The intercept (10.227) signifies that when the distance is zero, meaning the fire occurs right next to the fire station, the expected fire damage is about \$10,228. In our opinion, this intercept has no real-world meaning, as a fire happening directly at a fire station is unusual.

C. Correlation Test & Visualization

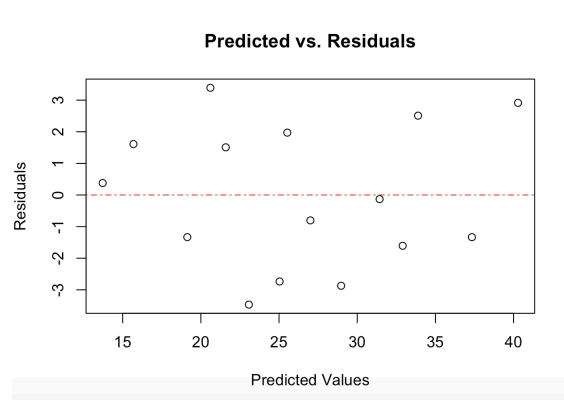
To further validate the relationship, a correlation test was performed, and a scatter plot was created. The Pearson Correlation Coefficient between Distance and Damage was 0.961, indicating a very strong positive linear relationship. This reinforces the regression analysis results, confirming that as the distance from the fire station increases, damage also tends to increase. When looking at the p-value of the test, it was extremely small, meaning the correlation is statistically significant.



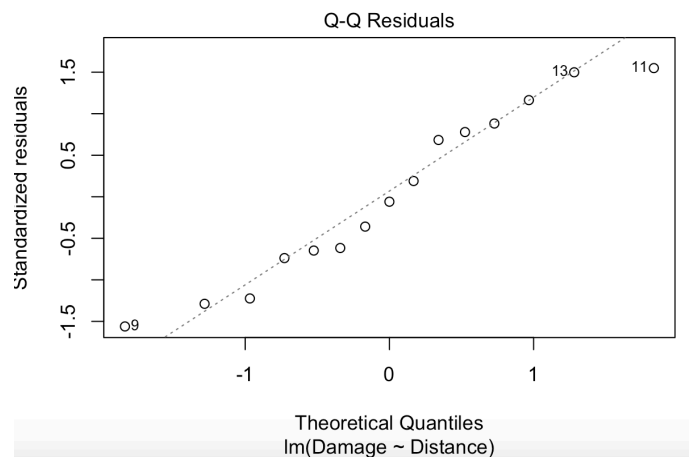
A scatterplot of the two variables showed a strong positive linear relationship, with most data points closely clustered around the fitted regression line. This visually confirms the strong correlation coefficient ($r = 0.96$) and suggests that the model effectively captures the association between the two variables.

D. Model Assumptions

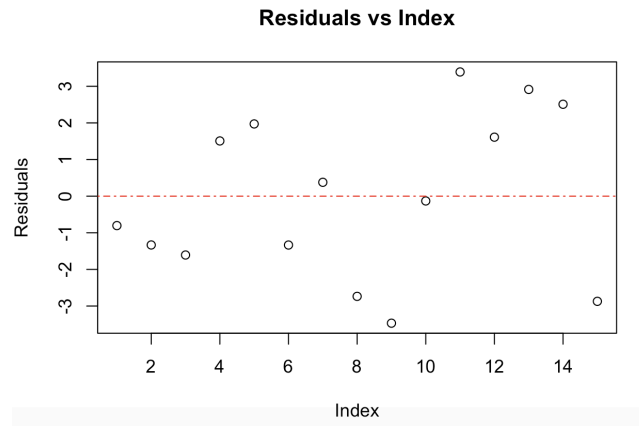
In order to ensure the validity and reliability of our model conclusions, we need to check the model assumptions. These model assumptions are the Linearity, Equal Variance, Independence, and Normal Population Assumptions.



When looking at the Predicted vs. Residuals plot of our regression model, you can see that there is no violation of Linearity and Equal Variance as the points are randomly spread around 0. There is also an equal spread of the points, as there appears to be no clustering and an absence of any sort of pattern.



In order to check for the Normal population assumption, we need to observe the Normal Probability Plot of residuals for the regression model. As you can see by the plot above, it follows a straight line, although not perfectly; therefore, there is no violation of Normality.



Finally, to check for the Independence assumption, we need to observe the Residuals vs Index plot of the regression model. When looking at the plot, you can see that the points are randomly scattered around zero, suggesting that the residuals are independent. There are also no signs of any sort of patterns or trends. Based on these observations, it's safe to conclude that there are no violations of Independence. After seeing that all the model assumptions were met, we can now ensure our model conclusions are valid and safe to assume.

III. Study 2

A. Model Summary & Interpretation

For Study 2, I examined the factors influencing executive compensation by modeling the natural logarithm of salary(y) as the response variable. In this analysis, we investigate the association between the response variable and the 10 potential predictor variables using multiple regression techniques and variable selection methods.

There are 10 potential predictor variables:

x₁: Experience — Years of professional experience (**Quantitative**)

x₂: Education — Total years of formal education (**Quantitative**)

x₃: Bonus Eligibility — Whether the individual is eligible for a bonus (1 = Yes, 0 = No)
(**Qualitative**)

x₄: Employees Supervised — Number of employees currently supervised (**Quantitative**)

x₅: Corporate Assets — Value of corporate assets in millions of dollars (**Quantitative**)

x₆: Board Member — Whether the individual is a board member (1 = Yes, 0 = No)
(**Qualitative**)

x₇: Age — Age of the individual in years (**Quantitative**)

x₈: Company Profits — Company profits over the past 12 months, in millions of dollars
(**Quantitative**)

x₉: International Responsibility — 1 if it has international responsibilities, 0 otherwise
(**Qualitative**)

x₁₀: Company Sales — Total sales in the past 12 months (in millions of dollars) (**Quantitative**)

To prepare the data for regression analysis, I converted the qualitative variables, **X3** (Bonus Eligibility), **X6** (Board Membership), and **X9** (International Responsibility) into factors using the factor function in R. This step is essential because multiple linear regression models require categorical variables to be treated as factors. Since the variables were qualitative, we needed to change this so R can know to treat them as categorical and generate the appropriate dummy variables. This helps improve the accuracy of the model, allowing for meaningful interpretation and the coefficients associated with those predictors.

Summary Model 1:

```

Residuals:
      Min       1Q   Median       3Q      Max
-0.201770 -0.050464  0.004435  0.046826  0.185952

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.002e+01  1.481e-01  67.692 < 2e-16 ***
X1           2.792e-02  1.773e-03  15.745 < 2e-16 ***
X2           2.903e-02  3.426e-03   8.475 4.57e-13 ***
X31          2.243e-01  1.708e-02  13.135 < 2e-16 ***
X4           5.140e-04  4.922e-05  10.443 < 2e-16 ***
X5           2.048e-03  5.250e-04   3.901 0.000186 ***
X61          -1.538e-02  1.686e-02  -0.912 0.364124
X7           -5.097e-04  1.438e-03  -0.355 0.723795
X8           -2.633e-03  5.128e-03  -0.513 0.608896
X91          -2.656e-02  2.037e-02  -1.304 0.195613
X10          -9.774e-04  2.959e-03  -0.330 0.741955
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07608 on 89 degrees of freedom
Multiple R-squared:  0.9229,    Adjusted R-squared:  0.9142
F-statistic: 106.5 on 10 and 89 DF,  p-value: < 2.2e-16

```

Note: The variables appear as X31, X61, X91, which were originally qualitative variables and were then converted into factors (binary dummy variables), that now appear like this in the model output.

The model output showed an R-squared of 92.29%, meaning that 92.29% of the variability in the natural logarithm of executive salary can be explained by the 10 predictor variables included in the model. This indicates a very strong model fit based on the R-squared. Since we are looking at multiple predictors, we also need to consider the adjusted R-squared, which was 91.42%. This is a slightly lower value, but these two are very close, which still suggests that while the model explains a high proportion of the variance, and confirms that most predictors contribute meaningfully. In addition, many of the predictors, such as experience (X1), education (X2), bonus eligibility (X31), number of employees supervised (X4), and corporate

assets (X5), have very low p-values less than 0.001, indicating that they are statistically significant and that there is strong evidence of a real relationship between these variables and executive salary.

B. Regression Equation

As the model met the key indicators of a strong fit, I formed the following equation:

$$\text{Salary} = 10.02 + 0.0279 \text{ X Experience} + 0.0290 \text{ X Education} + 0.2243 \text{ X Bonus Eligibility} + 0.000514 \text{ X Employees Supervised} + 0.00205 \text{ X Corporate Assets} - 0.0154 \text{ X Board Member} - 0.00051 \text{ X Age} - 0.00263 \text{ X Company Profits} - 0.0266 \text{ X International Responsibility} - 0.00098 \text{ X Company Sales}$$

This equation shows that some variables have strong positive relationships with salary. Firstly, the intercept is 10.02 in the regression equation, but it is not very meaningful because it is unrealistic to have no experience or education, sales, profits, etc.. Now, looking at the variables, the coefficient for Experience (0.0279) means that for every additional year of professional experience, an executive's expected log-salary increases by approximately 2.79%. The coefficient for Education (0.0290) implies that for every additional year of formal education, there's a 2.90% increase in expected salary. It turns out that Bonus Eligibility has a strong effect on executive salary, where being eligible for a bonus adds 0.2243 to the log-salary, which is close to about a 23% increase in actual salary. This suggests that bonus eligibility is a strong indicator of higher executive compensation.

The variables, Corporate Assets (0.00263) and Employees Supervised (0.000514), are statistically significant, though their coefficients are very small. On the other hand, variables Board Membership (X6), Age (X7), Company Profits (X8), International Responsibility (X9),

and Company Total Sales (X10) showed no statistically significant relationship with salary in this model. Not only were their coefficient small, but their p-values were greater than the threshold of 0.05.

C. Model Selection Techniques

Summary Model 2:

```

Residuals:
    Min       1Q   Median       3Q      Max
-0.201219 -0.056016 -0.003581  0.053656  0.187251

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  9.9619345   0.1010567   98.578 < 2e-16 ***
x1            0.0272762   0.0010293   26.501 < 2e-16 ***
x2            0.0290921   0.0033367    8.719 9.71e-14 ***
x31           0.2246932   0.0163503   13.742 < 2e-16 ***
x4            0.0005244   0.0000474   11.064 < 2e-16 ***
x5            0.0019623   0.0004972    3.947 0.000153 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07512 on 94 degrees of freedom
Multiple R-squared:  0.9206,    Adjusted R-squared:  0.9164
F-statistic: 218.1 on 5 and 94 DF,  p-value: < 2.2e-16

```

To improve the model, backward selection was used since we needed to address the multiple non-significant predictors. After this process, the model retained only 5 predictors: Experience (X1), Education (X2), Bonus Eligibility (X31), Employees Supervised (X4), and Corporate Assets (X5). These variables all had very low p-values, less than 0.001, indicating they are statistically significant contributors to explaining log-salary. The new Adjusted R^2 is 91.64%, maintaining a strong fit, and the residual standard error improved slightly, which suggests tighter predictions. Overall, backward selection helped improve the model by

10)

eliminating the predictors that were not significant to the original model, and leaving the most significant predictors.

Summary Model 3:

```
Residuals:
    Min       1Q   Median       3Q      Max
-0.201219 -0.056016 -0.003581  0.053656  0.187251

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  9.9619345   0.1010567   98.578 < 2e-16 ***
X1            0.0272762   0.0010293   26.501 < 2e-16 ***
X31           0.2246932   0.0163503   13.742 < 2e-16 ***
X4            0.0005244   0.0000474    11.064 < 2e-16 ***
X2            0.0290921   0.0033367    8.719 9.71e-14 ***
X5            0.0019623   0.0004972    3.947 0.000153 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07512 on 94 degrees of freedom
Multiple R-squared:  0.9206,    Adjusted R-squared:  0.9164
F-statistic: 218.1 on 5 and 94 DF,  p-value: < 2.2e-16
```

In addition to the backward selection, to ensure the most optimal model, forward selection was used. It turns out that it is identical to the backward selection model. It matched the adjusted R^2 and the residual standard error, and it included the same five significant predictors (Experience (X1), Education (X2), Bonus Eligibility (X31), Employees Supervised (X4), and Corporate Assets (X5)). So both methods, forward and backward selection, produced the same model, and forward selection serves as additional validation that these predictors are the most relevant for explaining differences in executive salary.

Summary Model 4:

```

Residuals:
    Min       1Q   Median       3Q      Max
-0.201219 -0.056016 -0.003581  0.053656  0.187251

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  9.9619345   0.1010567   98.578 < 2e-16 ***
X1           0.0272762   0.0010293   26.501 < 2e-16 ***
X31          0.2246932   0.0163503   13.742 < 2e-16 ***
X4           0.0005244   0.0000474    11.064 < 2e-16 ***
X2           0.0290921   0.0033367    8.719 9.71e-14 ***
X5           0.0019623   0.0004972    3.947 0.000153 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07512 on 94 degrees of freedom
Multiple R-squared:  0.9206,    Adjusted R-squared:  0.9164
F-statistic: 218.1 on 5 and 94 DF,  p-value: < 2.2e-16

```

To further refine the model, forward stepwise selection was applied, which combines both elements of forward and backward selection by adding predictors one at a time. This resulted in the same optimal model as forward and backward selection, with the same five significant predictors, and no change or loss compared to the full model, which means it confirms model stability, and that this is the most optimal model.

D. Multicollinearity

Since the forward stepwise, backward, and forward selection models all resulted in the same five predictors as the most statistically significant, I decided to focus on these variables from now on. To ensure the validity of the model, I checked for the presence of multicollinearity among the predictors therefore, a Variance Inflation Factor (VIF) analysis was conducted, and the VIF values presented below were as follows:

	X1	X2	X3	X4	X5
	1.002071	1.037777	1.063135	1.101590	1.029408

12)

(

As you can see, there is no evidence of multicollinearity since the VIF values were close to 1, meaning that there is no correlation among the predictor variables and the variables are completely independent of the other predictors, therefore, the model's predictions are accurate.

To ensure further absence of multicollinearity, a correlation matrix was used to examine whether there were any significant relations between the predictors. The correlation matrix was generated using only the significant quantitative variables: Experience (X1), Education (X2), Employees Supervised (X4), and Corporate Assets (X5). The correlation matrix is shown as follows:

	X1	X2	X4	X5
X1	1.0000000000	0.0005013485	-0.01021038	0.004393161
X2	0.0005013485	1.0000000000	-0.17930640	0.045454379
X4	-0.0102103795	-0.1793063990	1.00000000	0.107647806
X5	0.0043931614	0.0454543787	0.10764781	1.000000000

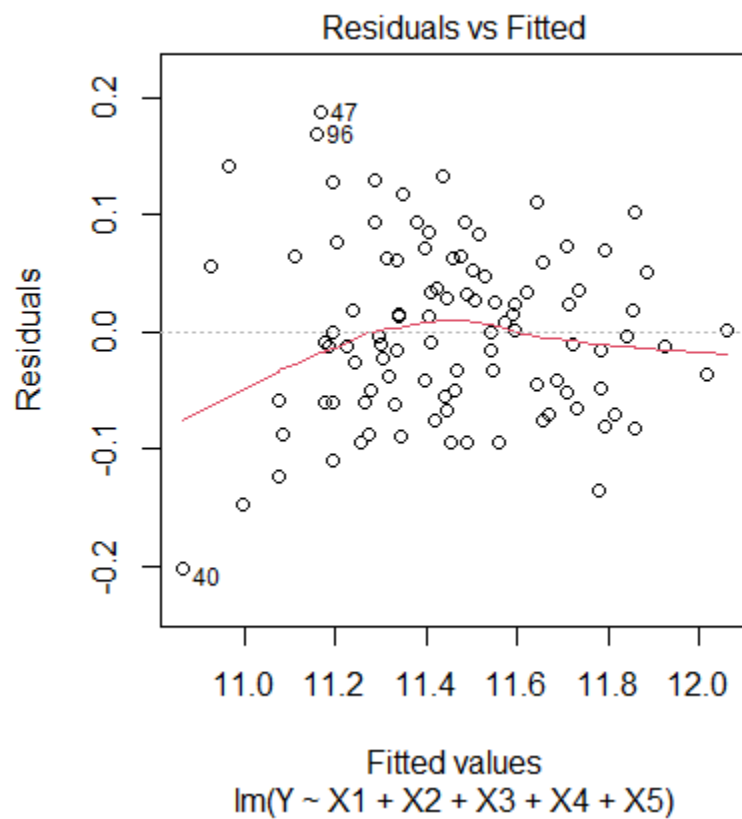
It shows that the correlation coefficients were all low, with the highest being between Education (X2) and Employees Supervised (X4) at -0.179, so it means that the predictors are not strongly linearly related to one another and that multicollinearity is not a concern for the model.

13)

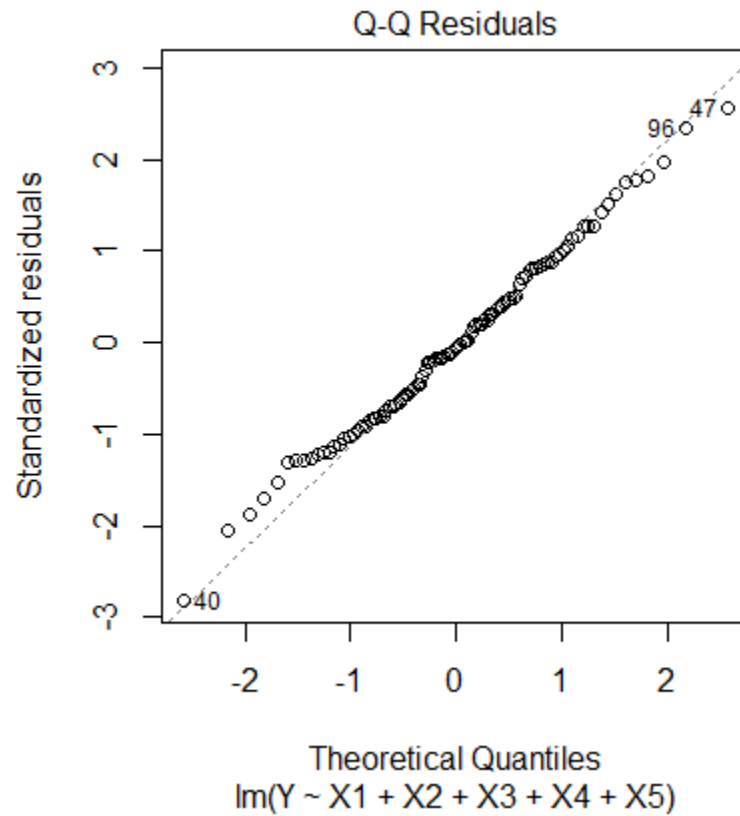
(

E. Model Assumptions

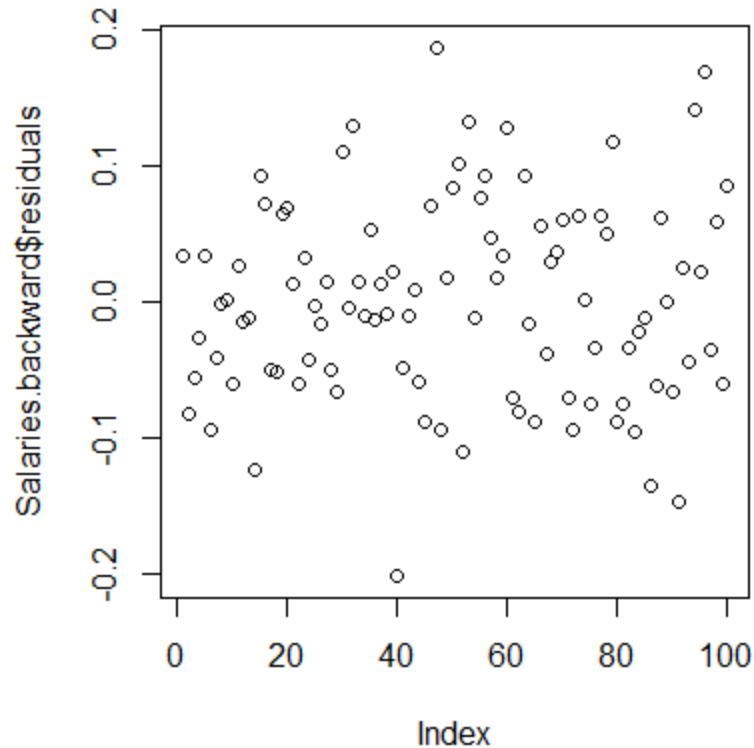
It is essential to check the model assumptions to ensure validity and reliability for our model conclusions, so firstly, we assess both linear and equal variance assumptions of the regression model.



Looking at the Residuals vs Fitted Values plot, the residuals appear randomly scattered around zero, the spread of residuals is relatively constant, and there is no noticeable curve, so there is no violation of linearity and equal variance.



This is the Q-Q plot for checking the normality assumption, where we examined the plot of standardized residuals. The points follow the diagonal line closely, and it means that the residuals are normally distributed, even though there are a few points that are slightly off the line; the pattern is still strong for the normality of the residuals, which satisfies this assumption for regression.



This is a Residuals vs Index plot, and it is used to check the independence assumption. Based on the plot of residuals, they appeared randomly distributed around zero with no patterns or trend over time, indicating that the residuals are independent of one another, so there's no violation of independence.

IV. Conclusion

Overall, by applying a simple linear regression model in Study 1 and a multiple regression model in Study 2, I successfully identified the significant predictors in each scenario. In Study 1, I found that *distance* (measured in miles) is a significant predictor of *fire damage* (measured in dollars), with greater distances from a fire station associated with increased damage. In Study 2, after utilizing various model selection techniques to refine our analysis, I determined that *years of experience*, *years of education*, *bonus eligibility*, *number of employees*

16)

(

supervised, and *corporate assets* are significant predictors of salary. These findings highlight the importance of thoughtful model-building in uncovering relationships among variables to make meaningful insights.